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Evaluating the Effects of Soil and Water Conservation Measures on Soil Moisture Regimes and Crop Performance in Jammu's Rainfed Agroecosystems

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Graphical abstract

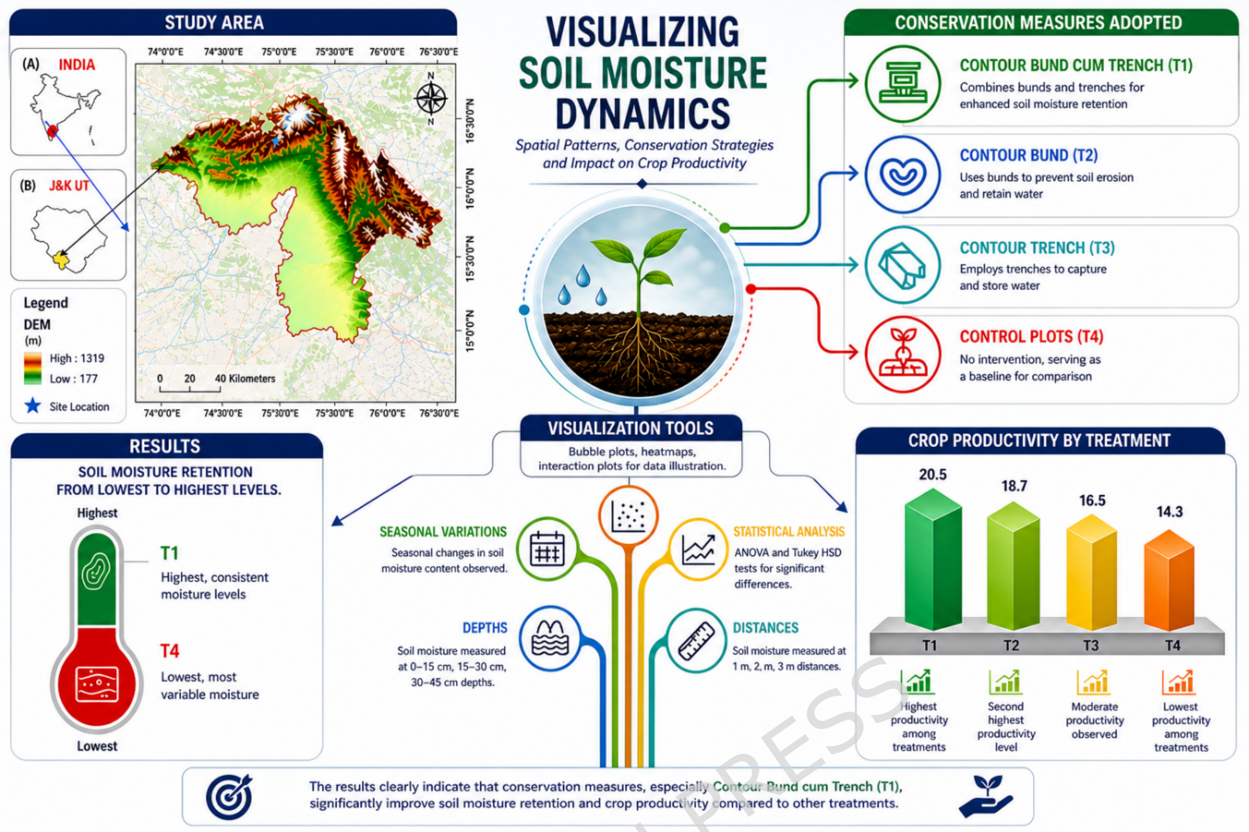
STUDY AREA
(A) INDIA
(B) J&K UT
Legend
DEM
(m)
High : 1319
Low : 177
Site Location
0 20 40 Kilometers
74°0'0"E
74°30'0"E
75°0'0"E
75°30'0"E
76°0'0"E
76°30'0"E
16°30'0"N
16°0'0"N
15°30'0"N
15°0'0"N
N
VISUALIZING SOIL MOISTURE DYNAMICS
Spatial Patterns, Conservation Strategies and Impact on Crop Productivity
CONSERVATION MEASURES ADOPTED
CONTOUR BUND CUM TRENCH (T1)
Combines bunds and trenches for enhanced soil moisture retention
CONTOUR BUND (T2)
Uses bunds to prevent soil erosion and retain water
CONTOUR TRENCH (T3)
Employs trenches to capture and store water
CONTROL PLOTS (T4)
No intervention, serving as a baseline for comparison
VISUALIZATION TOOLS
Bubble plots, heatmaps, interaction plots for data illustration.
RESULTS
SOIL MOISTURE RETENTION FROM LOWEST TO HIGHEST LEVELS.
Highest
T1
Highest, consistent moisture levels
T4
Lowest, most variable moisture
Lowest
SEASONAL VARIATIONS
Seasonal changes in soil moisture content observed.
DEPTHS
Soil moisture measured at 0–15 cm, 15–30 cm, 30–45 cm depths.
STATISTICAL ANALYSIS
ANOVA and Tukey HSD tests for significant differences.
DISTANCES
Soil moisture measured at 1 m, 2 m, 3 m distances.
CROP PRODUCTIVITY BY TREATMENT
20.5
18.7
16.5
14.3
T1
T2
T3
T4
Highest productivity among treatments
Second highest productivity level
Moderate productivity observed
Lowest productivity among treatments
The results clearly indicate that conservation measures, especially Contour Bund cum Trench (T1), significantly improve soil moisture retention and crop productivity compared to other treatments.

Figure 1: Graphical representation of soil moisture dynamics and crop productivity under different conservation measures.

Abstract

Sustainable soil moisture management is critical for rainfed hill agriculture in the Kandi region of Jammu. This study evaluated four soil and water conservation (SWC) treatments—contour bund cum trench (T1), contour bund (T2), contour trench (T3), and untreated control (T4)—to assess their effects on soil moisture across depth, distance from conservation structures, and time, and to examine the associated response of wheat productivity under rainfed conditions. The field study was conducted during the rabi season using wheat (*Triticum aestivum* L.) variety JAUW 598. Soil moisture was monitored at three depths (0–15, 15–30, and 30–45 cm) and three distances (1, 2, and 3 m) from the conservation structures during the study period. Statistical analysis showed that soil moisture was significantly affected by conservation treatment, soil depth, distance from the conservation structures, and sampling time ($p < 0.001$ for all main effects). Among the evaluated treatments, T1 consistently recorded the highest soil moisture, typically around 7.8–8.0%, whereas T2 and T3 showed intermediate values (about 7.0–7.5%) and T4 recorded the lowest values (often below 6.6%). Soil moisture was generally higher in the 0–15 cm layer and declined with increasing distance from the conservation structures. Improved moisture conservation was also reflected in crop response, with wheat grain yield increasing by 43.3% under T1, 30.7% under T3, and 15.3% under T2 relative to the untreated control. These site-specific findings suggest that the contour bund cum trench system was the most effective treatment for improving soil moisture in the upper 0–45 cm profile and supporting wheat productivity under the rainfed conditions of the study area.

Keywords: Soil and water conservation; Soil moisture dynamics; Contour bund cum trench; Rainfed wheat; Shiwalik foothills; Spatio-temporal variability; Soil moisture retention; Jammu agroecosystem

1. Introduction

Water resources are fundamental to ecological stability, agricultural production, and sustainable development. Globally, agriculture accounts for the largest share of freshwater use (69%), followed by industry (19%) and domestic consumption (12%), while overall demand continues to rise because of population growth, urbanization, climate variability, and inefficient resource management (UNESCO, 2018; Chen et al., 2023; Salehi, 2022). Global population is projected to increase from 8.0 billion in 2023 to about 9.4–10.2 billion by 2050, potentially exposing 2.7–3.2 billion people to severe water scarcity (Musie and Gonfa, 2023). Agriculture is central to this challenge because irrigation alone accounts for more than 60% of global freshwater withdrawals, directly linking water availability with food security (Kang et al., 2017). At the same time, land resources are under increasing pressure from competing development priorities, including food production, water security, climate action, and land restoration under the Sustainable

Development Goals. Expansion of cultivated land to meet projected food, feed, and fiber demand—expected to rise by nearly 60% by 2050—often accelerates deforestation, biodiversity loss, and land degradation, highlighting the need for integrated land and water management approaches (Singh et al., 2018; Kumawat et al., 2023; Yadav et al., 2023).

Soil degradation, particularly soil erosion, remains one of the most serious threats to land productivity and ecosystem sustainability. Although erosion is a natural geomorphic process, accelerated erosion caused by deforestation, overgrazing, shifting cultivation, and unsustainable farming practices results in substantial losses of fertile topsoil and nutrients (Kumar et al., 2021; Singh et al., 2023). In sloping cultivated lands, soil loss may exceed 15–30 Mg ha^{-1} yr^{-1} compared with less than 5 Mg ha^{-1} yr^{-1} under stable forest cover (Arunrat et al., 2023). Beyond reducing soil fertility and crop productivity, eroded sediments contribute to downstream siltation, flooding, and contamination of water bodies. In India, annual soil loss is estimated at 5.3 billion tonnes, corresponding to an economic loss of about ₹89,000 crore. Water erosion is the dominant degradation process, affecting nearly 83 million hectares, while wind erosion further contributes to deterioration, together impacting more than 95 million hectares with an average soil loss rate of 16.23 t ha^{-1} (Mandal et al., 2021; Singh et al., 2023).

These concerns are especially severe in rainfed agriculture, which occupies nearly 80% of global cultivated land and contributes 60–70% of food production, yet remains highly vulnerable to rainfall variability and poor soil-moisture retention (Li et al., 2018). In fragile mountain ecosystems such as the Himalayan foothills, climate variability intensifies these constraints through erratic precipitation, reduced infiltration, rapid runoff, and severe erosion (Körner et al., 2019; Peters et al., 2019; Schmeller et al., 2022). The Shiwalik Hills are among the most erosion-prone landscapes, characterized by steep slopes, fragile soils, low organic carbon, and highly variable rainfall, resulting in severe gully development and chronic water scarcity (Sharma et al., 2009; Kukal and Bhatt, 2006; Arora and Bhatt, 2016). These conditions strongly constrain agricultural productivity and livelihood security in rainfed farming communities. Under such conditions, soil and water conservation (SWC) interventions are essential for improving infiltration, reducing runoff, stabilizing soil moisture, and enhancing crop productivity. Mechanical measures such as contour bunding, trenching, vegetative barriers, and integrated land shaping have shown promise in reducing erosion and improving water-use efficiency in sloping terrains (Ranjan et al., 2017; Qin et al., 2015; Li et al., 2017). However, their performance often varies with rainfall regime, soil properties, slope gradient, and management intensity, making location-specific evaluation necessary for identifying robust strategies suited to fragile hill agroecosystems.

Despite the reported benefits of SWC practices in sloping rainfed regions, field-based evidence remains limited on how individual and combined conservation measures influence soil moisture simultaneously across soil depth, lateral distance from conservation structures, and time under the specific agro-ecological conditions of the Shiwalik foothills of Jammu. In particular, there is inadequate location-specific information on whether the interaction of conservation treatment with soil depth and field position alters spatio-temporal moisture distribution in ways that are agronomically meaningful for rainfed cropping systems. Therefore, the present study evaluated four SWC treatments—contour bund cum trench (T1), contour bund (T2), contour trench (T3), and untreated control (T4)—to quantify their effects on soil moisture across different depths, distances, and observation periods, and to assess the associated response of wheat productivity under rainfed conditions. We hypothesized that the integrated contour bund cum trench system would conserve higher and more stable soil moisture across the soil profile and field transect than the individual measures or the control, thereby improving crop performance. The novelty of this study lies in its integrated field-based evaluation of treatment × depth × distance × time effects on soil moisture in a rainfed Shiwalik agroecosystem, generating location-specific evidence on how conservation structures influence both the magnitude and spatial distribution of water availability within the crop root zone.

2. Materials and Methods

2.1 Study area

The study was conducted in Kandi village (Figure 2), Akhnoor Tehsil, Jammu district, India, located at 32°59′34″ N and 74°40′55″ E, at an elevation of 773 m above sea level. The village lies approximately 20 km from Akhnoor and 60 km from Jammu and covers about 563 ha. The area has a subtropical climate and represents a typical rainfed hill agroecosystem of the Kandi tract. Long-term average annual rainfall is 1154.52 mm (2003–2017), of which about 74% is received during June to September. Air temperature generally ranges from about 4 °C in winter to 40 °C in summer. Agriculture in the Kandi area is predominantly rainfed, and farm ponds serve as an important local source of supplemental water (Directorate of Economics and Statistics, 2015; Census 2011).

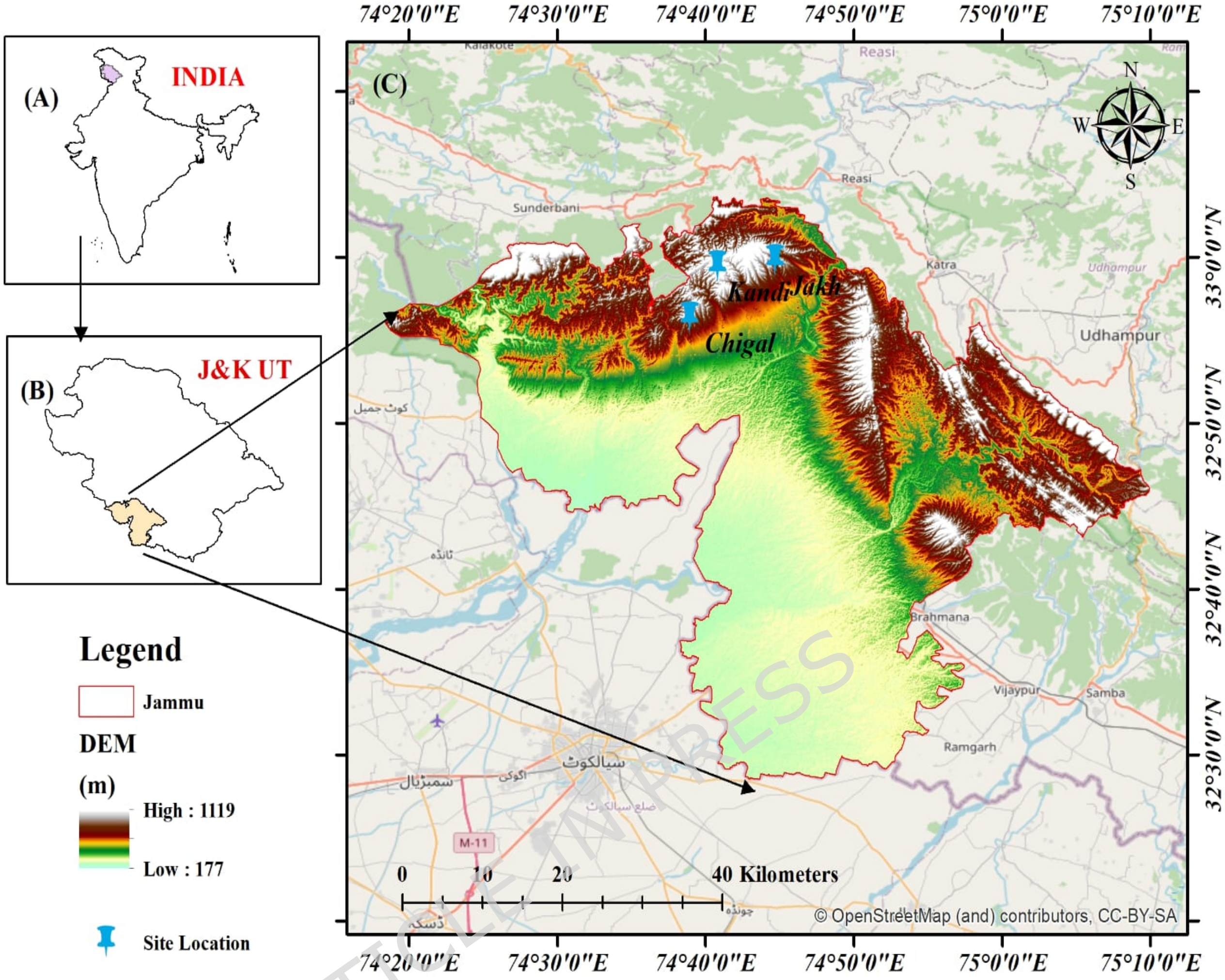
(A)
INDIA
(B)
J&K UT
(C)
Legend
Jammu
DEM
(m)
High : 1119
Low : 177
Site Location
Kandi
Jakh
Chigal
N
W
E
S
74°20'0"E
74°30'0"E
74°40'0"E
74°50'0"E
75°0'0"E
75°10'0"E
33°0'0"N
32°50'0"N
32°40'0"N
32°30'0"N
0
10
20
40 Kilometers
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Figure 2: Geographic map of the study area highlighting the locations of various sites where soil and water conservation measures were implemented. The map was generated using software ArcGis(10.8) USA (http://www.esri.com/arcgis).

2.2 Experimental design and layout

The study was conducted as a replicated comparative field evaluation under rainfed conditions in Kandi village during the rabi season using wheat (*Triticum aestivum* L.) variety JAUW 598. Four field conditions were evaluated: contour bund cum trench (T1), contour bund (T2), contour trench (T3), and an untreated control (T4). Each treatment was represented by four independent field plots serving as replicates, with each plot measuring approximately 20 m × 5 m (100 m²).

The plots were selected from adjacent areas with broadly comparable slope position, soil condition, and land-use history to enable direct comparison between conserved and non-conserved conditions. Because the conservation structures had already been established under field conditions, the study was conducted as a comparative field evaluation rather than as a newly imposed completely randomized trial. Within each replicate plot, soil moisture observations were recorded at three soil depths and three horizontal distances from the conservation structures. Corresponding sampling positions were maintained in the control plots for comparison. Thus, the replicate field plot was considered the experimental unit, while soil depth and distance from conservation structure were treated as within-plot sampling factors. The experimental factors and coded levels used for soil-moisture and grain-yield assessment are summarized in Table 1.

Table 1. Experimental factors and coding used in the study

Factor	**Code**	**Description**
Soil and water conservation measure	T1	Contour bund cum trench
	T2	Contour bund
	T3	Contour trench
	T4	Untreated control
Soil depth (cm)	D1	0–15
	D2	15–30
	D3	30–45
Distance from conservation structure (m)	L1	1
	L2	2
	L3	3

Factor	Code	Description
Soil and water conservation measure	T1	Contour bund cum trench
	T2	Contour bund
	T3	Contour trench
	T4	Untreated control

Factor	Code	Description
Soil depth (cm)	D1	0–15
	D2	15–30
	D3	30–45
Distance from conservation structure (m)	L1	1
	L2	2
	L3	3

2.3 Crop and agronomic management

The experiment was conducted using wheat (*Triticum aestivum* L.) variety JAUW 598 during the rabi (winter) season under rainfed conditions. The crop was raised following the recommended agronomic practices for rainfed wheat in the region, and all crop-management operations, including land preparation, sowing, fertilizer application, weed management, and plant-protection measures, were kept uniform across all treatments. This ensured that differences in soil moisture dynamics and crop productivity could be attributed primarily to the soil and water conservation measures rather than to variation in crop management.

2.4 Soil sampling and moisture determination

Soil samples were collected using a soil auger from three depth intervals: 0–15 cm, 15–30 cm, and 30–45 cm below the surface. Samples were obtained at three distances from the conservation structures: 1 m (L1), 2 m (L2), and 3 m (L3). Each sample was placed in a labeled polythene bag for proper identification and handling, following Burt (2009). To capture the soil layer most directly influenced by rainfall infiltration, runoff interception, and the effects of contour-based soil and water conservation structures, soil moisture was monitored within the 0–45 cm profile at 0–15, 15–30, and 30–45 cm intervals. This depth range was selected to represent the upper soil profile and much of the active root zone of the wheat crop under rainfed conditions, where short-term changes in plant-available water are expected to respond most directly to conservation measures. However, soil moisture below 45 cm was not measured; therefore, the present analysis reflects upper-profile soil-moisture dynamics and does not fully capture deeper water reserves that may contribute to crop water uptake during later growth stages.

Fresh weight was recorded immediately after collection. The samples were then oven-dried at 105 °C for 24 h to constant mass, and oven-dry weight was determined using an analytical balance. Gravimetric soil moisture content was calculated using Equation (1):

$$\textbf{Soil moisture (\%)} = \frac{Wf\text{-}Wd}{Wd} \times 100 \quad 1$$

where Wf is the fresh weight of the soil sample (g) and Wd is the oven-dry weight of the soil sample (g). Soil textural composition (sand, silt, and clay) was determined using the Bouyoucos hydrometer method

2.5 Grain-yield measurement

Grain yield was determined at physiological maturity by harvesting the crop from five randomly placed 1 m^2 quadrats within each replicate plot. The harvested produce was dried, weighed, and converted to grain yield expressed as q ha^{-1}. The mean of the five quadrats was used as the plot-level grain-yield observation for statistical analysis.

2.6 Statistical analysis

Because soil moisture was measured repeatedly over time within the same plots and sampling positions, the soil-moisture data were analyzed using linear mixed-effects models to account for the hierarchical and repeated-measures structure of the experiment. Conservation treatment (T1–T4), soil depth (D1–D3), distance from conservation structure (L1–L3), and sampling time were treated as fixed effects, including their relevant interactions. Replicate plot was included as a random effect, and repeated observations from the same sampling positions over time were modeled within the mixed-effects framework to account for non-independence. Pairwise comparisons among fixed-effect means were performed using Tukey-adjusted estimated marginal means at the 5% significance level. Model assumptions were evaluated using residual-versus-fitted plots and normal Q–Q plots of residuals to assess homoscedasticity and normality. To improve interpretability, effect sizes were summarized using marginal and conditional R^2 values for the fitted mixed models. Grain yield data were summarized as mean ± standard deviation (SD) on a plot basis and analyzed using one-way analysis of variance (ANOVA) with treatment as the main factor, followed by Tukey's Honest Significant Difference (HSD) test at $p < 0.05$. To quantify the agronomic linkage between moisture conservation and crop response, a linear regression analysis was performed between plot-level mean seasonal soil moisture and grain yield. Statistical analyses were carried out using Python (version 3.10) and relevant packages.

3. Results

3.1 Soil moisture variation across conservation treatments, depths, and distances

Soil moisture varied among conservation treatments, soil depths, and distances from the conservation structures (Figures 3 and 4). Across most treatment combinations, the contour bund cum trench treatment (T1) recorded the highest mean soil moisture, generally around 7.8–8.0%. Contour bund (T2) and contour trench (T3) showed intermediate values, mostly about 7.0–7.5%, whereas the untreated control (T4) recorded the lowest values, often below 6.6%. The bubble plot further indicated lower variability under T1 than under the other treatments, whereas T3 and T4 showed greater variation across treatment combinations. The boxplots showed a similar distributional pattern, with higher median moisture and narrower spread under T1 and lower median values with wider spread under T4.

Across the soil profile, moisture was generally highest in the 0–15 cm layer and declined with depth. However, T1 maintained comparatively higher moisture even in the 15–30 cm and 30–45 cm layers. Across field positions, soil moisture was highest at 1 m from the conservation structure and declined toward 2 m and 3 m, although T1 retained a moisture advantage even at the farthest distance.

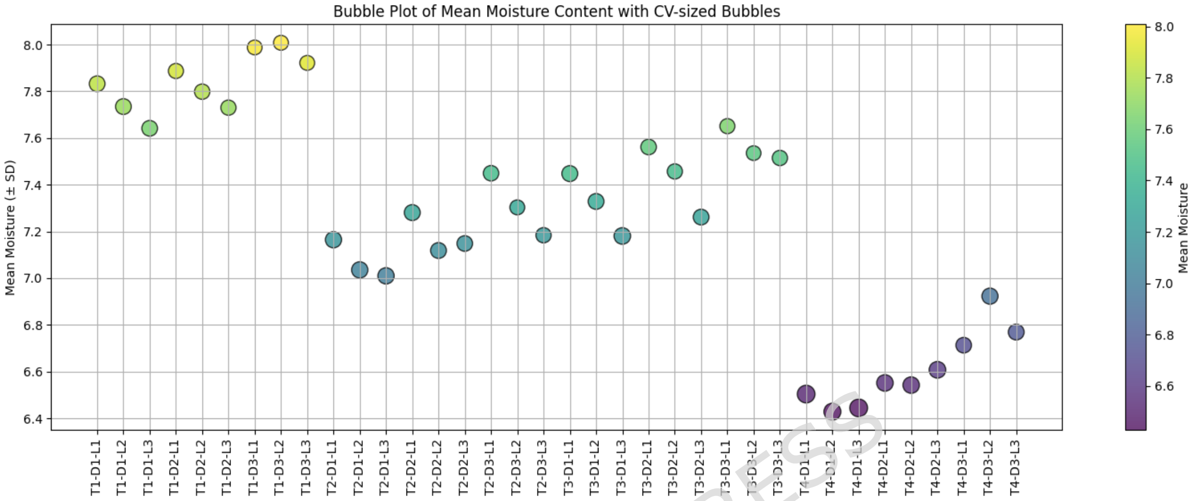

Figure 3. Bubble plot of mean soil moisture (±SD) under conservation measures (T1–T4), depths (D1–

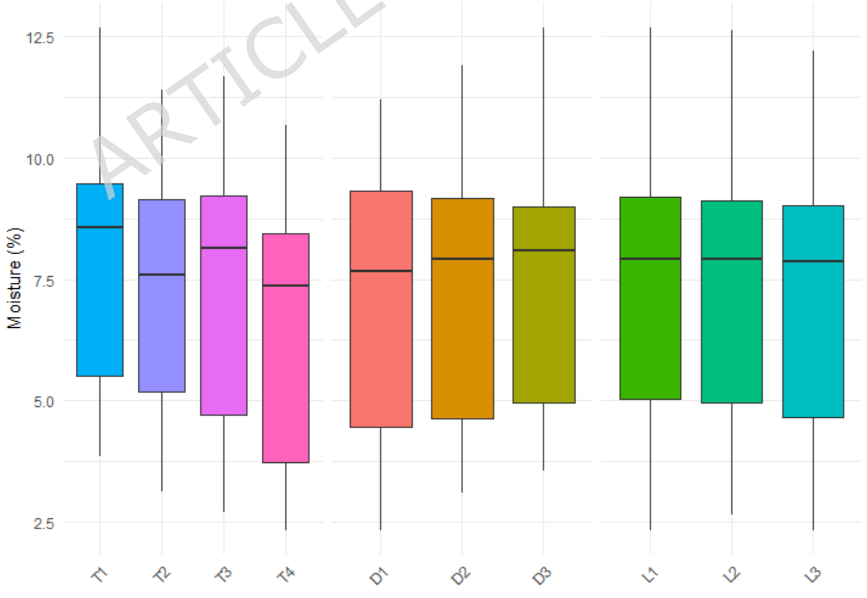

Figure 4. Distribution of soil moisture across conservation measures, soil depths, and distances from conservation structures.

3.2 Temporal variation in soil moisture

Temporal variation in soil moisture was evident across the observation period (Figures 5–7). Mean soil moisture increased from late autumn into winter, with the highest values observed during late December and early January, and another rise was evident in November. Lower values were recorded during April and October. Across most observation periods, T1 maintained higher moisture than T2, T3, and T4. In the treatment × depth plots, T1 at D1 and D2 frequently exceeded 10% during wetter periods, whereas T4 remained consistently lower and showed pronounced reductions at deeper depths during drier periods.

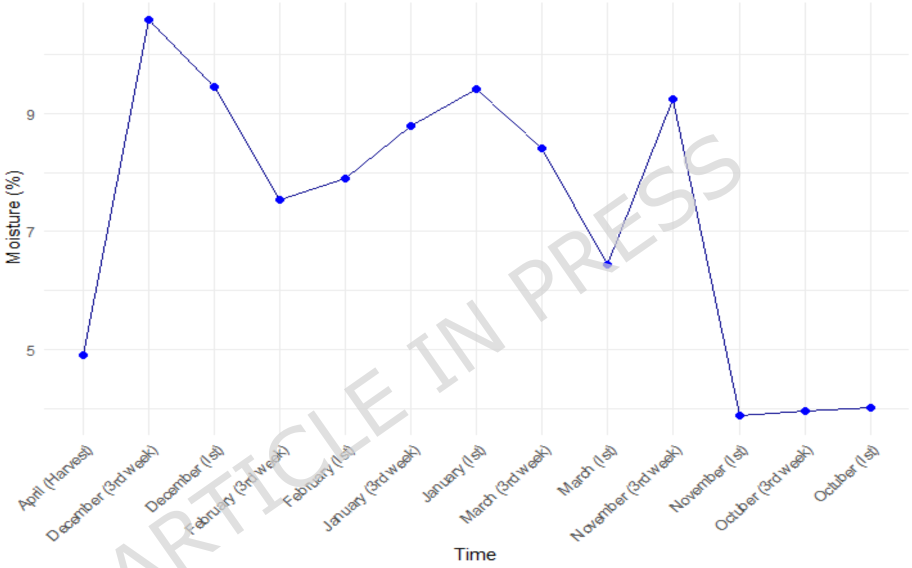

Figure 5. Moisture dynamics across months and weeks.

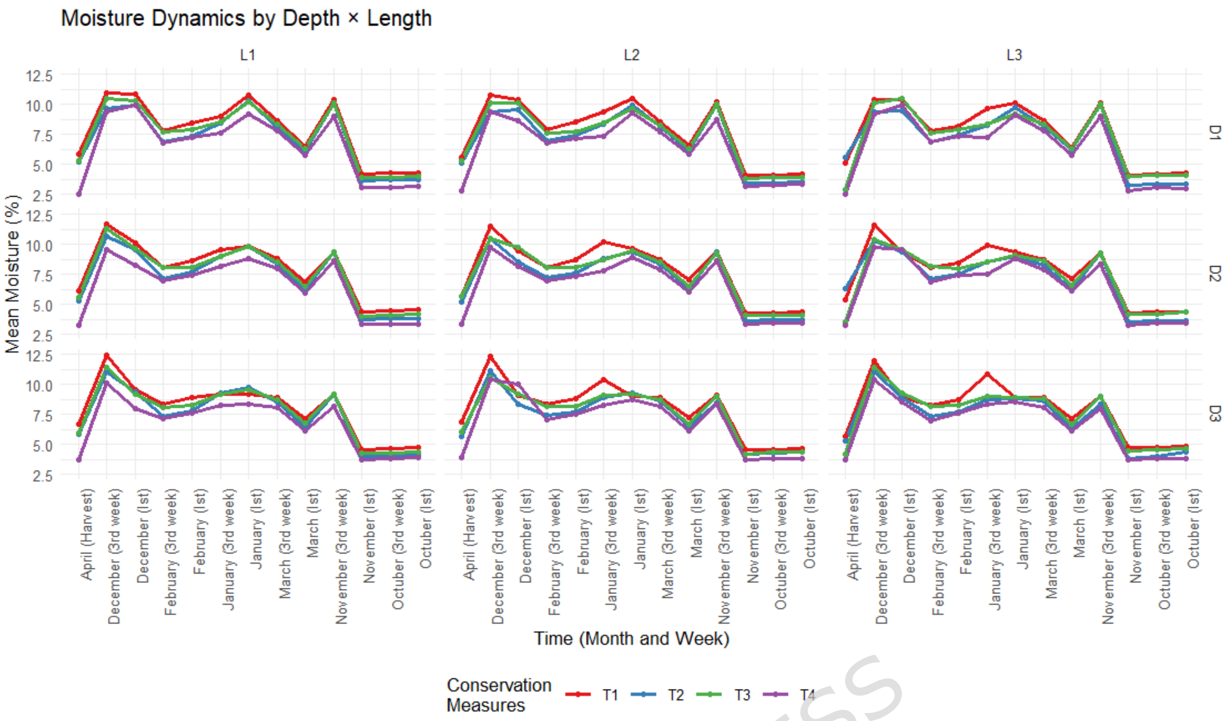

Figure 6: Soil Moisture Trends across treatments and depths over time

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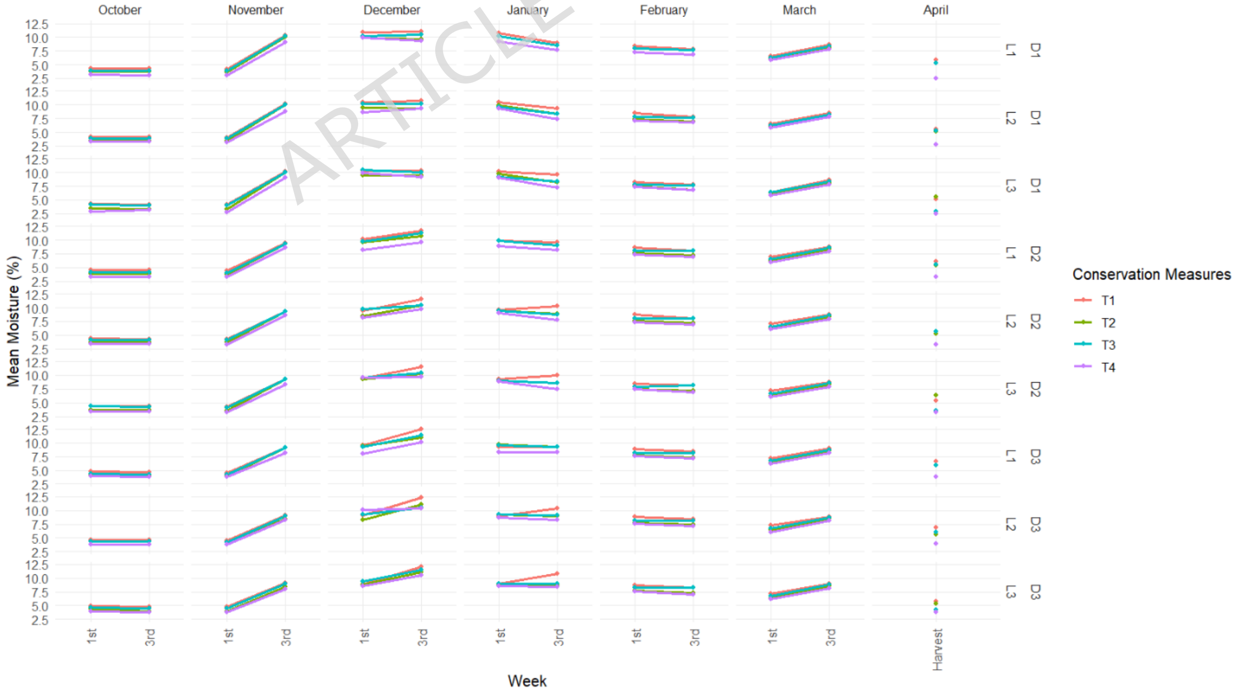

Figure 7: Soil moisture variation by treatment, depth, and distance over months and weeks

3.3 Statistical evidence for treatment, spatial, and temporal effects

Analysis of variance showed that soil moisture was significantly influenced by conservation measure ($F = 1975.84$, $p < 0.001$), depth ($F = 181.69$, $p < 0.001$), distance from the conservation structures ($F = 45.16$, $p < 0.001$), week ($F = 12241.37$, $p < 0.001$), and month ($F = 20352.30$, $p < 0.001$) (Table 2). Significant interaction effects were also observed for conservation measure × depth ($F = 2.16$, $p = 0.045$), conservation measure × distance ($F = 8.54$, $p < 0.001$), depth × distance ($F = 2.51$, $p = 0.040$), conservation measure × week ($F = 185.49$, $p < 0.001$), depth × week ($F = 69.58$, $p < 0.001$), distance × week ($F = 41.12$, $p < 0.001$), conservation measure × month ($F = 21.03$, $p < 0.001$), depth × month ($F = 39.57$, $p < 0.001$), distance × month ($F = 6.85$, $p < 0.001$), week × month ($F = 5788.49$, $p < 0.001$), and conservation measure × week × month ($F = 23.32$, $p < 0.001$). These interaction terms show that the magnitude of treatment differences was not constant across depth, distance, and time.

The heatmap supported these patterns by separating higher-moisture and lower-moisture combinations (Figure 8). Most T1 combinations were associated with the higher-moisture cluster, whereas most T4 combinations were grouped in the lower-moisture cluster. Combinations involving shallower depth and shorter distance were also more closely associated with the higher-moisture group.

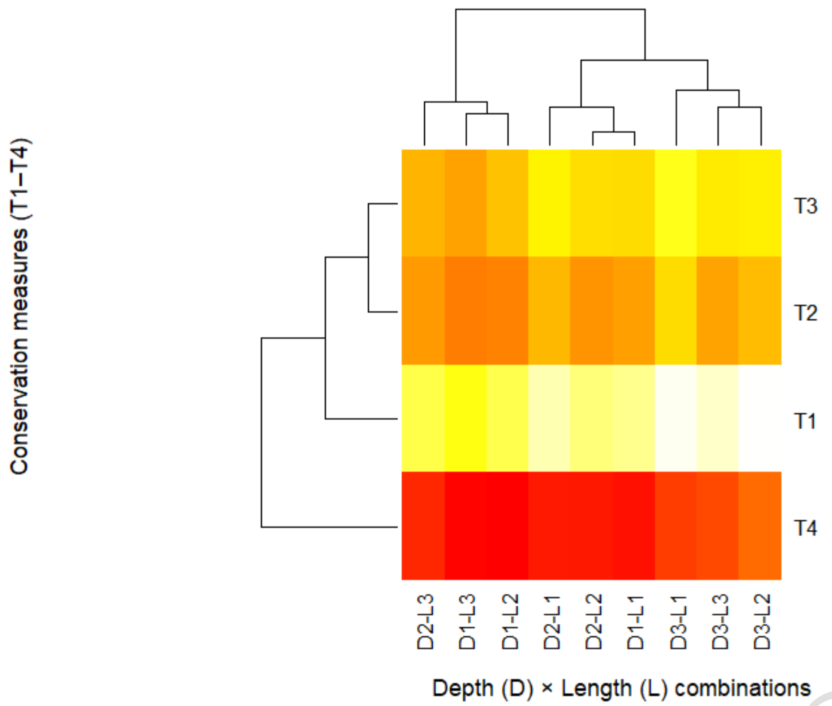

Figure 8: Heatmap of soil moisture under different conservation measures, soil depths, and distances from conservation structures.

3.4 Pairwise comparisons

Tukey’s HSD test showed significant pairwise differences among most conservation treatments (Figures 9–11). T1 differed significantly from T2, T3, and T4, whereas the difference between T2 and T3 was not significant. Significant contrasts were also observed among several depth, distance, and temporal comparisons, with shallower layers and shorter distances generally associated with higher soil moisture. Together, these pairwise comparisons were consistent with the patterns observed in the descriptive figures and Table 2.

Table 2: ANOVA Results for main effects and interactions influencing soil moisture dynamics

Source	Df	F-value	p-	Significan
Main Effects				
Conservation measures	3	1975.84	<	***

Depth	2	181.69	<	***
Length	2	45.16	<	***
Week	2	12241.3	<	***
Month	5	20352.3	<	***
Two-Way Interactions				
Conservation measures × Depth	6	2.16	0.045	*
Conservation measures × Length	6	8.54	<	***
Depth × Length	4	2.51	0.040	*
Conservation measures × Week	6	185.49	<	***
Depth × Week	4	69.58	<	***
Length × Week	4	41.12	<	***
Three-Way Interactions				
Conservation measures × Month	1	21.03	<	***
Depth × Month	1	39.57	<	***
Length × Month	1	6.85	<	***
Week × Month	5	5788.49	<	***
Higher-Order Effects				
Conservation measures × Week ×	1	23.32	<	***

*Significance codes: *** p < 0.001; ** p < 0.01; * p < 0.05; . p < 0.1; ns*

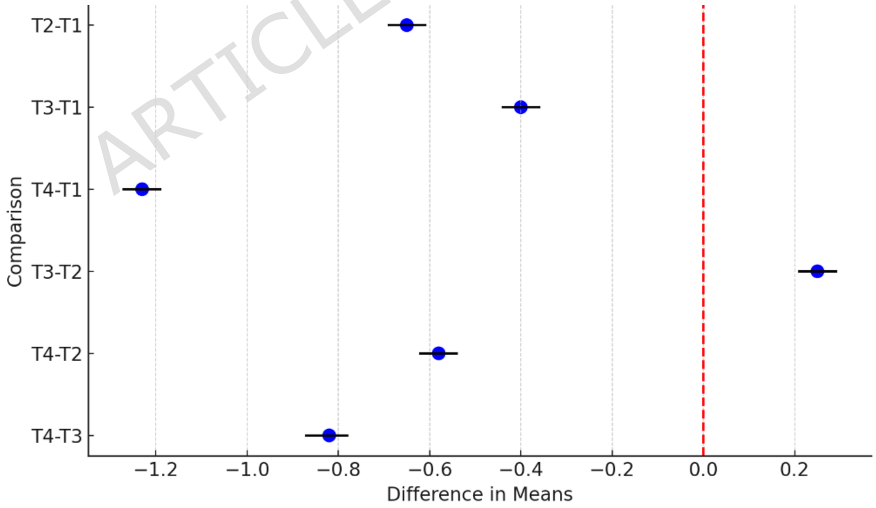

Figure 9. Tukey HSD pairwise comparisons among conservation measures (T1–T4) for soil moisture content.

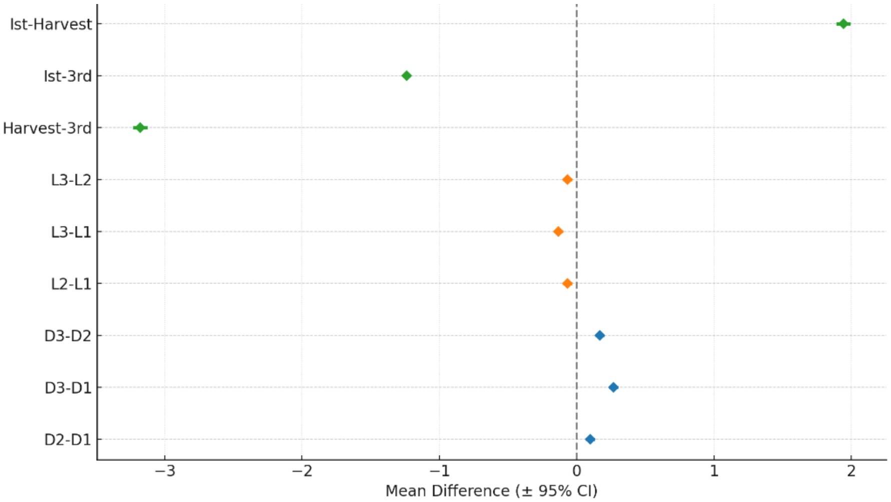

Figure 10. Tukey HSD comparisons of main effects for Depth (D1–D3), Length (L1–L3), and Weeks (Harvest to 3rd week).

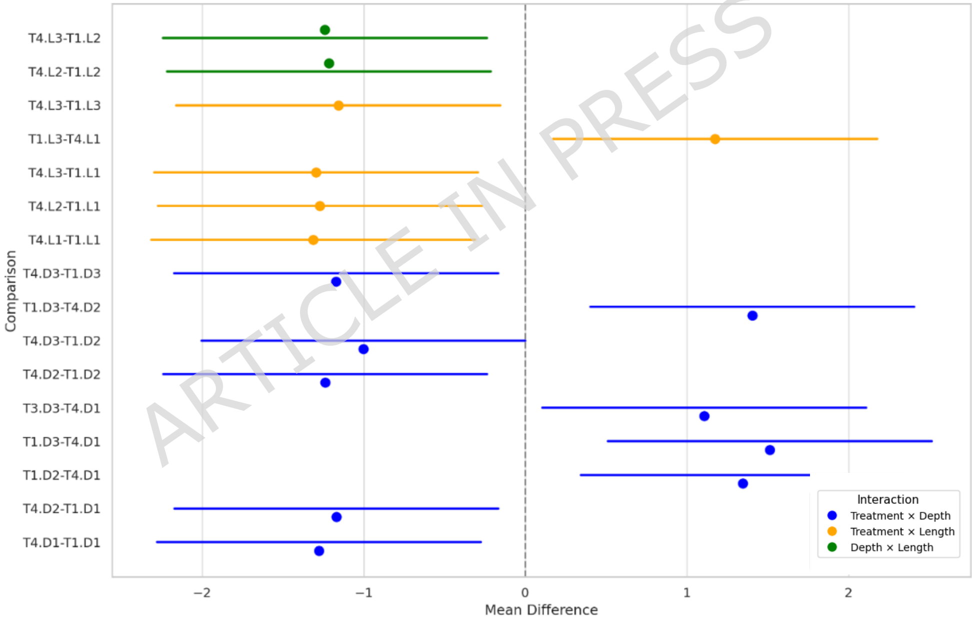

Figure 11. Tukey HSD comparisons for significant interactions among interactions

3.5 Grain yield response under different conservation measures

Grain yield response under different conservation measures: Grain yield as shown in figures 12 and 13 differed significantly among the conservation treatments (ANOVA: $F = 59.89$, $p < 0.001$). The contour bund cum trench treatment (T1) recorded the highest wheat grain yield (31.53 ± 1.36 q ha^{-1}), followed by contour trench (T3: 28.75 ± 1.17 q ha⁻¹), contour bund (T2: 25.37 ± 0.92 q ha⁻¹), and the untreated control (T4: 22.00 ± 0.71 q ha⁻¹). Relative to the control, grain yield increased by 43.3% under T1, 30.7% under T3, and 15.3% under T2. Tukey's HSD test showed that T1 was significantly higher than T3, T2, and T4; T3 was significantly higher than T2 and T4; and T2 was significantly higher than T4. A positive relationship was observed between grain yield and mean seasonal soil moisture ($R^2 = 0.90$, $p < 0.001$), indicating that treatments with greater retained soil moisture also produced higher wheat yield, with T1 showing both the highest moisture status and the highest grain yield.

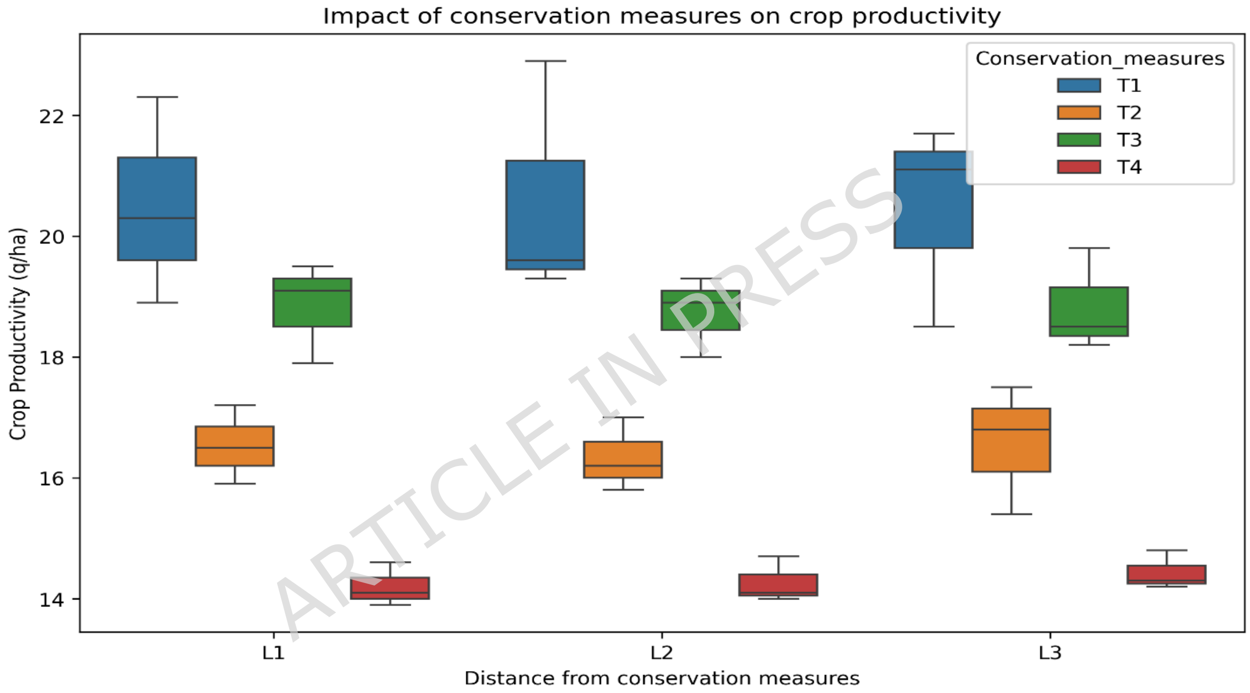

Fig 12**.** Boxplot showing the effect of different conservation measures on wheat grain yield

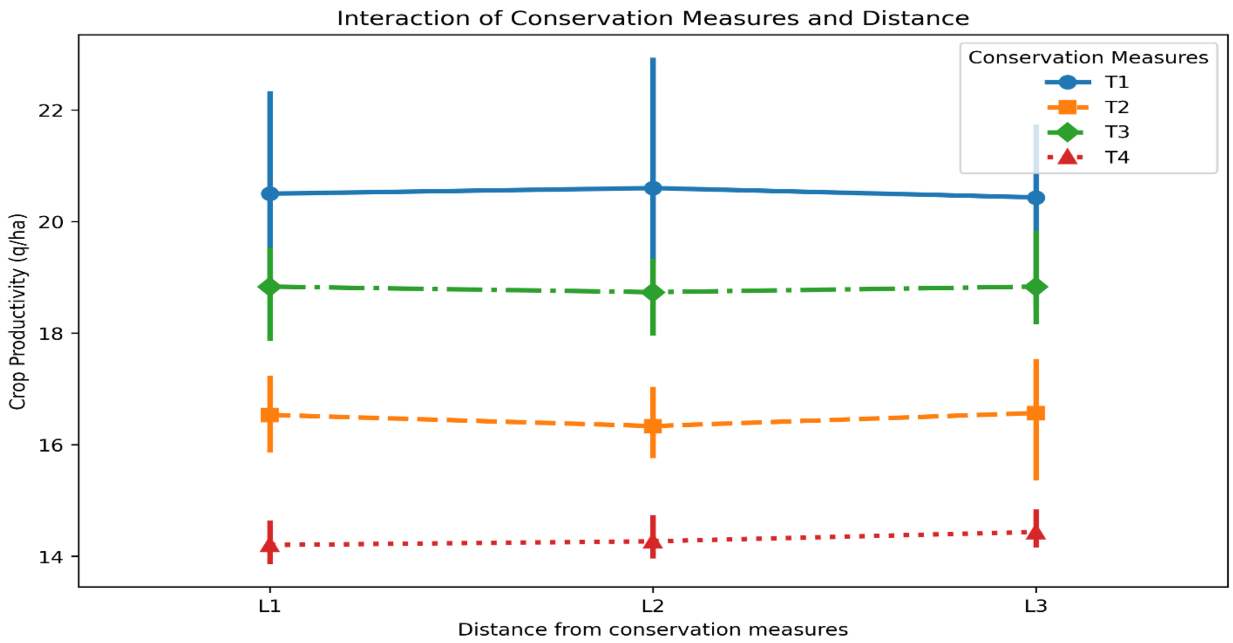

Fig 13: Interaction plot showing the combined effects of treatment and distance from conservation structures on wheat grain yield.

4. Discussion

The present study demonstrates that soil and water conservation (SWC) measures altered both the magnitude and persistence of soil moisture under rainfed conditions in the Shiwalik foothills of Jammu, with clear differences among treatments. Across depth, distance, and time, the contour bund cum trench system (T1) consistently maintained the highest soil moisture, whereas the untreated control (T4) recorded the lowest values. This pattern indicates that the integrated system was more effective than the individual measures in conserving water within the upper soil profile, while contour bund (T2) and contour trench (T3) provided intermediate benefits. Similar advantages of combined SWC measures over single interventions have been reported in sloping and semi-arid environments, where integrated structures improve water retention and reduce erosion more effectively than standalone bunding or trenching (Birhanu et al., 2022; Ghosh et al., 2021; Fantahun et al., 2024; Fikadu et al., 2022). The treatment differences can be interpreted within a basic soil-water balance framework. Under rainfed sloping conditions, rainfall is partitioned into runoff, infiltration, temporary soil-water storage, evapotranspiration, and drainage. The contour bund in T1 likely reduced slope length and runoff velocity, thereby increasing the opportunity time for infiltration, while the trench intercepted and temporarily stored overland flow, allowing more water to infiltrate vertically into the soil profile rather than being lost rapidly as surface runoff. This combined effect explains why T1 maintained higher soil moisture than T2, T3, and T4 across most depth, distance, and time combinations. The comparatively weaker performance of T2 and T3 suggests that runoff control alone or trenching alone provided only partial hydrological benefit, whereas the integrated bund-and-trench configuration improved both runoff reduction and infiltration opportunity (Pendhare et al., 2021; Panigrahi et al., 2009; Kumar et al., 2021). The spatial pattern of soil moisture further supports this interpretation. Moisture was generally higher at 1 m from the conservation structures and declined toward 2 m and 3 m, indicating localized interception and redistribution of runoff. However, the persistence of the T1 advantage even at 3 m suggests that the integrated system improved not only immediate water capture but also lateral redistribution and storage within the field. Likewise, higher moisture in the 0–15 cm layer across treatments reflects the direct influence of infiltration near the soil surface, whereas the sustained T1 advantage at 15–30 and 30–45 cm indicates greater downward movement of water into the measured profile. In contrast, the sharper decline with depth and distance under T4 indicates that, without conservation measures, a larger fraction of rainfall was likely lost as runoff and shallow evaporation, leaving less water available within the root zone. Comparable spatial benefits of SWC practices have been reported in other rainfed watersheds, where conservation structures altered soil-moisture patterns beyond their immediate footprint (Damtie et al., 2022; Retta et al., 2022). Temporal variation remained strongly influenced by seasonal conditions, with higher soil moisture during wetter and cooler periods and lower values during drier intervals. Although rainfall and evapotranspiration were not explicitly incorporated into the present analysis, the smaller seasonal decline under T1 suggests a moisture-buffering effect of the integrated structure. By increasing infiltration and short-term storage during wetter periods, T1 likely slowed the rate of soil-water depletion during subsequent dry spells. This buffering function is particularly important in rainfed wheat systems, where short-term moisture stress during establishment and reproductive stages can strongly limit productivity. Similar resilience benefits of SWC measures under variable seasonal conditions have been described in semi-arid hill and watershed environments (Ghosh et al., 2021; Birhanu et al., 2022; Fantahun et al., 2024).

Soil physical properties likely also influenced the magnitude of treatment effects. Soil texture governs water intake and storage, while bulk density affects pore continuity, infiltration, and root-zone penetration. Soils with lower bulk density and more favorable pore structure generally allow greater infiltration and storage, whereas denser soils restrict downward movement and increase the likelihood of runoff and near-surface drying. Thus, the moisture patterns observed here were likely shaped not only by the conservation structures themselves but also by the soil physical environment in which those structures operated. Integrating texture, bulk density, and infiltration behavior therefore provides a mechanistic basis for understanding why T1 produced the most stable soil-water regime. The grain-yield response followed the same general pattern as soil-moisture conservation. The highest wheat yield under T1 and the lower yields under T2, T3, and especially T4 indicate that improved soil-water availability was translated into better crop performance under rainfed conditions. By maintaining greater moisture within the upper 0–45 cm profile during critical growth periods, the contour bund cum trench system likely reduced moisture stress and supported more favorable crop growth. Thus, the agronomic advantage of T1 appears to be functionally linked to its superior moisture-conservation performance rather than to any difference in crop management, which was kept uniform across treatments. The findings should, however, be interpreted within the scope of the study. The evaluation was conducted at a single location during one cropping season, and soil moisture was monitored only up to 45 cm depth. Therefore, the results are most directly applicable to the upper soil profile and to agroecological conditions similar to those of the rainfed Shiwalik foothills of Jammu. In addition, climatic drivers such as rainfall and evapotranspiration were not quantitatively linked to the moisture observations, and broader hydrological responses such as runoff and groundwater recharge were not measured. Multi-location and multi-season studies, ideally integrating climate variables, deeper soil-water storage, runoff measurements, and quantitative moisture–yield relationships, would strengthen broader inference and help refine recommendations for wider application.

5. Conclusion

This study unequivocally demonstrates that integrated soil and water conservation measures, particularly the contour bund cum trench system (T1), substantially enhance soil moisture retention across diverse spatial and temporal scales in the rainfed, semi-arid landscapes of Jammu’s Shiwalik foothills. Compared to single measures like contour bunds (T2) and trenches (T3), T1 consistently maintained higher mean soil moisture, sustained moisture into deeper soil layers, and extended benefits horizontally across the field, even at distances up to 3 meters from the conservation structures. Moreover, T1 effectively buffered seasonal moisture fluctuations, preserving soil water during critical dry periods post-harvest and between monsoon cycles, while also reducing spatial variability and ensuring more uniform moisture distribution. These insights not only corroborate previous findings from semi-arid regions globally but also highlight the specific advantages of combined bund-and-trench systems in hilly agro-ecosystems where slope, rainfall variability, and soil heterogeneity pose significant challenges. The significant interactive effects observed among conservation measures, depth, distance, and seasonal factors underscore the complex but manageable dynamics governing soil moisture under conservation interventions. The increase in crop productivity in the contour bund cum trench was found to be 43.3% in comparison to the control. In the case of contour bund and contour trench, the increase in crop productivity in comparison with the control plot was 15.3% and 30.7% respectively. Collectively, our findings advocate for prioritizing integrated SWC practices like contour bund cum trench systems as a cornerstone strategy for enhancing moisture security and stabilizing rainfed agriculture in vulnerable hill regions. Such measures hold the promise of transforming water-limited environments into more resilient and productive landscapes, offering a sustainable pathway to safeguard livelihoods against the growing threats of climate variability and water scarcity. Nonetheless, future research should extend these findings across multiple cropping cycles, diverse topographies, and socio-economic contexts, while integrating direct crop yield assessments and broader hydrological impacts to fully quantify the long-term benefits of these conservation strategies.

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Declarations

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